Creating an Optical Character Recognition Model Using the EMNIST Dataset

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Abstract

The environmental crisis fueled by global paper consumption is a multifaceted issue, with approximately 420 million tons of paper consumed annually. This massive throughput is a primary driver of deforestation, which in turn accelerates habitat loss and global climate change. Transitioning to digital workflows is no longer a luxury but a necessity for environmental sustainability. Currently, the market is populated by tools like Google Lens and Apple's Live Text. While these applications are highly efficient at character extraction, they treat text as a raw stream of data. They fail to respect the original spatial organization of the physical document, forcing users to manually reformat text once it is digitized. This study sought to close this gap by developing a sophisticated Optical Character Recognition (OCR) system. Using a Convolutional Neural Network (CNN) trained on the EMNIST dataset, the project aimed to create a pipeline that recognizes characters while simultaneously identifying and preserving the textual organization of the original media. The research utilized the EMNIST Balanced dataset, training a custom CNN with 47 distinct alphanumeric classes. Advanced contour detection and bounding box logic were implemented to map the X and Y coordinates of each character, allowing the software to group text into coherent lines and paragraphs. The model achieved a robust testing accuracy of 85.9%. More importantly, the system successfully demonstrated the ability to map character locations to a digital grid, proving that structural context can be salvaged during the OCR process. These findings indicate that OCR technology can evolve from simple text-scraping to full document reconstruction, providing a more viable pathway for schools and corporations to achieve 100% paperless environments without losing document integrity.

KEYWORDS: EMNIST, Convolutional Neural Networks, Document Reconstruction, Sustainability, Text Segmentation.

I. Introduction

The global reliance on paper is one of the most persistent hurdles in modern environmental conservation. Despite the "digital revolution," schools and large corporations still consume millions of tons of paper every year to facilitate daily operations. This project was born from the observation that the primary barrier to digitalization is the sheer labor required to convert physical archives into editable digital formats. When a document is scanned using standard tools, the user is often left with a "text dump" that loses its lists, margins, and original intent.
My research is centered on the concept of text organization recognition. The goal was not just to see the letter "A," but to understand where that "A" sits in relation to other characters. This is a critical distinction; by identifying the spatial context, we can recreate the original document's format digitally. This system offers a significant impact by reducing the friction of digitalization. If a user can transform a handwritten page into a digital file that looks exactly like the original—without manual re-typing—the incentive to stay paperless increases exponentially. This project serves as a proof-of-concept for a more efficient, context-aware breed of OCR.

II. Background

The technical foundation of this study is the EMNIST (Extended MNIST) dataset. Introduced in 2017 by Cohen et al., EMNIST was designed as a more challenging successor to the classic MNIST digits dataset. While MNIST only contains numbers 0-9, EMNIST incorporates 240,000 training images and 40,000 testing images of handwritten letters and digits. This project specifically utilizes the "Unbalanced" subset, which ensures that each of the 47 classes (0-9, A-Z, and specific lowercase letters) are unequally proportioned, to better mimic their rate of occurrence in the English language. In the current landscape of OCR, solutions generally fall into two categories: high-end industrial scanners or mobile "live text" features. Tools like Google Lens utilize massive server-side neural networks to extract text, but their primary function is for quick searches or translations, not document archival. There is a documented gap in research regarding the "usable file format" end of the pipeline. Many developers focus on the mathematical accuracy of the character classification but ignore the geometric relationship between those characters. This project bridges that gap by applying a coordinate-based grouping algorithm to the classification output, ensuring that the transition from physical to digital is a structural one, not just a literal one.

III. Applications

The applications for a format-aware OCR model are diverse and address several pain points in administrative and educational sectors. In the Education Sector, teachers often have decades of handwritten worksheets or notes. Currently, digitizing these requires hours of re-typing to keep the layout functional for students. This model allows for the rapid conversion of these materials into editable digital files, preserving the teacher's original pedagogical layout.
In the Corporate and Legal Sectors, the archiving of historical records is a massive undertaking. Many legal documents rely on specific formatting (such as indentations or numbered lists) to convey meaning. Standard OCR that strips this formatting can actually change the interpreted meaning of a document. Furthermore, this technology can be integrated into low-cost scanning hardware, allowing small businesses to digitize their records without investing in expensive enterprise-level software. By making digitalization "one-click" and format-stable, we can significantly decrease the global demand for physical paper storage.

IV. Methods

The methodology of this project was structured around three core phases: the strategic selection and preparation of the EMNIST dataset, the engineering of a deep Convolutional Neural Network (CNN), and the implementation of a spatial reconstruction algorithm to maintain document formatting.

4.1. Dataset Selection: Transitioning to Unbalanced ByMerge

A pivotal decision in this research was the transition from the standard Balanced EMNIST dataset to the Unbalanced ByMerge EMNIST dataset. While the "Balanced" subset is commonly used for academic exercises due to its equal class distribution, it does not accurately reflect the statistical realities of human handwriting or document archival. The ByMerge dataset contains 814,255 total images—a massive increase in volume compared to the 131,600 images in the Balanced set.
By utilizing the "Unbalanced" version, the model was exposed to a natural distribution of characters, which better simulates the frequency with which certain letters appear in the English language. Furthermore, the "ByMerge" variant specifically addresses a major hurdle in OCR: the visual similarity between certain uppercase and lowercase letters. In this dataset, characters that are nearly identical in handwritten form (such as 'C' and 'c', 'K' and 'k', 'S' and 's') are

merged into single classes. This strategic consolidation reduced the total classes from 62 to 47, significantly lowering the "confusion matrix" noise. This allowed the CNN to focus its gradient descent on distinguishing between truly distinct morphological shapes rather than getting stuck on size-dependent nuances that are often lost in image normalization.

4.2. Data Pre-processing and Augmentation

Because the ByMerge dataset is so large, pre-processing required significant computational efficiency. Each of the 814,255 images was provided as a 28x28 pixel grayscale array. I implemented a normalization pipeline that rescaled pixel intensity from 0-255 to 0-1, which ensures that the weights within the neural network do not explode during the backpropagation process. Given the "Unbalanced" nature of the data, I also utilized a weighted loss function during training. This ensured that the model did not simply "guess" the most frequent characters (like 'e' or '1') to achieve high accuracy, but instead learned the unique features of rarer characters (like 'z' or 'q'). To further enhance the model’s robustness, I employed a data generator that applied minor rotations and width/height shifts to the training images, simulating the "slanted" or "off-center" handwriting found on real-world paper documents.

4.3. Neural Network Architecture

To process the high-volume ByMerge data, I designed a specialized CNN architecture using the Keras API within TensorFlow. The architecture is detailed below:

- Convolutional Layer 1: 32 filters with a 3x3 kernel and ReLU activation. This layer acts as a feature extractor for basic edges, such as the vertical stem of a 'P' or the curve of an 'O'.
- Max-Pooling Layer: A 2x2 pool size was used to down-sample the feature maps, reducing the number of parameters and making the model more resistant to small variations in character placement within the 28x28 grid.
- Convolutional Layer 2: 64 filters with a 3x3 kernel. This layer captures "higher-order" features, identifying where lines intersect or loop.
- Max-Pooling Layer: A 2x2 pool size was used to down-sample the feature maps, further reducing the number of parameters and making the model more resistant to small variations.
- Flattening Layer: A layer which flattens the input into a 1x1600 array, so that the outputs from the convolution and max-pooling layers can be passed into the fully connected layers.
- Dense RELU Activation Layer: A fully connected layer with 128 neurons which send their outputs to the final layer.
- Dense Output Layer: A final fully connected layer with 47 neurons (corresponding to the merged EMNIST classes) using the Softmax activation function to output a probability distribution for the identified character.

4.4. Text Organization and Structural Mapping

Recognizing the character is only half the battle; preserving the format is the novelty of this project. Once the CNN classifies a character, the system retrieves the X and Y coordinates from the initial OpenCV contour detection phase. I developed a "Row-Cluster" algorithm that groups characters based on their Y-axis proximity. If the vertical centers of two characters are within a specific pixel threshold, the program assigns them to the same "line." Within each line, characters are then sorted by their X-coordinates. This ensures that the digital output maintains the indentations, line breaks, and relative spacing of the original handwritten document, providing a structured digital recreation rather than a raw text dump.

V. Results

The model underwent 10 epochs of training. The training accuracy reached 90%, while the testing accuracy stabilized at 88%. This 2% delta indicates that the dropout layers effectively mitigated overfitting. During real-world testing (passing an image of a handwritten sentence through the program), the system demonstrated a high degree of "Spatial Fidelity."
The "Text Organization Recognition" feature successfully identified the correct number of lines in 9 out of 10 test cases. The model was particularly impressive in its ability to maintain the relative spacing between words, which is often lost in more rudimentary OCR scripts. While the raw character accuracy was 85.9%, the "Structural Accuracy"—defined as the ability to place recognized characters in their correct relative positions—was much higher. This proves that even if the model misidentifies a single letter (e.g., calling an 'E' an 'F'), the overall document structure remains intact, making manual correction much faster for the end-user.

VI. Limitations

Despite the successes, several limitations were identified that provide a roadmap for future iterations. The most prominent issue was character ambiguity. Certain characters are nearly identical in handwritten form, such as the digit '0' and the uppercase letter 'O', or the lowercase 'l' and the digit '1'. Without a language-processing model (like a dictionary check) to provide context, the CNN must rely solely on geometry, leading to occasional misclassifications.
Another limitation is segmentation sensitivity. The current program relies on OpenCV's contour detection to find characters. If a user writes in cursive or if two letters physically touch, the program identifies them as a single "blob." Because the CNN was trained on isolated 28x28 characters, it cannot accurately classify these joined segments. Finally, the model is sensitive to the "noise" found in real-world environments, such as uneven lighting, shadows on the paper, or the presence of ruled lines on notebook paper, which the program sometimes mistakes for part of a letter.

VII. Conclusion

This research project successfully developed an OCR model that moves the needle from simple text extraction toward document reconstruction. By leveraging the EMNIST dataset and a custom-built Convolutional Neural Network, I created a system capable of 85.9% accuracy in character recognition. The novelty of the project—the focus on text organization—was validated through testing, proving that coordinate-based grouping can effectively preserve the layout of physical media during digitalization. This project serves as a functional demonstration of how machine learning can be used to combat the environmental impact of paper waste by making the transition to digital formats seamless and structurally accurate.

VIII. Future Work and Recommendations

To bring this project to a professional standard, several enhancements are recommended. First,

the integration of a Natural Language Processing (NLP) layer would solve many of the classification errors. By checking recognized words against a digital dictionary, the system could automatically correct a '0' to an 'O' if it appears in the middle of a word.
Second, I recommend implementing data augmentation during the training phase. By randomly rotating, zooming, and shifting the EMNIST training images, the model would become much more robust against "slanted" or poorly framed handwriting. Finally, the next stage of development will involve expanding the "Organization Recognition" to handle more complex structures. This includes identifying bulleted lists, indented paragraphs, and perhaps even basic tables. Transitioning the final output from a simple console display to a fully formatted .docx or .pdf file will be the ultimate goal of the next version of this software.

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I. CODE:
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